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Topi, Khyber Pakhtunkhwa, Pakistan

MS-492 OPERATIONS MANAGEMENT

GIKAFE Cafeteria Operations Study

Forecasting, Inventory & Queueing Analysis at the Peak Lunch Window

Group Members

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1. Problem Statement

GIKAFE, the main on-campus cafeteria at GIK Institute, serves approximately 500 students through a single main counter during the 12:00–14:00 lunch window. Visible queues of 6–8 customers form at the peak; popular items such as Chicken Biryani and Naan occasionally run short before 14:00; and the cafeteria currently operates with no formal demand-forecasting method and no documented inventory policy. The problem studied here is twofold: **how to predict daily demand with usable accuracy**, and **how to size the order quantities and reorder points for the high-value items** that drive most of the cafeteria’s revenue.

2. Background and Context

GIKAFE is operated directly by GIK administration as a low-overhead service rather than a profit-maximising vendor. It runs seven days a week with one main service counter, a separate (informal) tea sub-counter, and a single cashier. The menu contains 16 items grouped into Main Meals, Beverages, Sides, Condiments, Bread, and Snacks, with unit prices ranging from PKR 20 (Naan, Chicken Nuggets) to PKR 250 (Chicken Karhai+Rice, Chicken Burger).

Daily footfall is heaviest on Fridays (Jummah prayer and post-prayer social lunch) and tapers off on Sundays (students travelling home for the week). The observed window was the peak lunch period of 12:00–14:00 across Friday 8-May, Saturday 9-May, and Sunday 10-May 2026, with a follow-up visit on Monday 11-May 2026 to confirm Day-11 actuals with the on-duty supervisor. Total observed footfall over the three observation days was **1,461 customers** (511 Fri / 483 Sat / 467 Sun).

3. System Boundary

Included	GIKAFE main service counter; 12:00–14:00 lunch window; 16 menu items; one server and one cashier; arrival and service-time observations at the main counter.
Excluded	Kitchen prep upstream of the counter; breakfast (07:00–10:00) and dinner (19:00–21:00) service; the informal tea sub-counter (not independently timed); supplier-side operations; cash handling and POS reconciliation.

The boundary is deliberately narrow — a single counter, a single peak window, a single menu — so that the data collected in the two-week study window is sufficient to support honest analysis rather than a broad but shallow review.

4. Reason for Selecting the Problem

Three reasons made this problem appropriate for a two-week applied study. First, the bottleneck is visible to the naked eye: anyone present at 12:30 can see customers queueing for 6–8 deep, which makes data collection straightforward and the recommendation intuitively credible. Second, the cafeteria has no existing forecasting or inventory policy, so even a moderate analytical improvement creates real value rather than competing with an existing system. Third, all three problem dimensions — demand variability, ordering policy, and peak congestion — map directly to standard OM tools covered in Heizer & Render (Chapters 4, 12, and D), making the project both methodologically clean and pedagogically aligned with the course syllabus.

5. Tools Selected and Justification

The project applies **two OM tools** as permitted by the project guidelines, with a third (queueing) used as supporting diagnostic evidence rather than a primary analytical instrument.

Tool	Source (H&R)	Why this tool fits
Forecasting — SMA, WMA, ES	Chapter 4	Cafeteria demand varies day-to-day but the historical record is short (10 days available). Short-horizon smoothing methods are the textbook fit for this data scale, and MAPE provides an objective basis to pick between them.
Inventory — ABC, EOQ, ROP	Chapter 12	With 16 items of very different unit values, ABC is the natural first step before any ordering policy is sized. EOQ and ROP then provide the standard trade-off between ordering cost and holding cost for the Class-A items that drive the majority of annual value.
Supporting: M/M/1 queueing	Module D	The 6–8 person queue is itself a measurable phenomenon. M/M/1 is used here only to confirm whether the single-server system is stable at the observed peak rate — it does not drive the primary recommendation.

The project guidelines explicitly state that “one tool applied well is better than two applied poorly.” We took this seriously: the forecasting and inventory tools are applied in full with all calculations shown, while the queueing model is reported only as a single-page diagnostic.

6. Data Collection Method

Two students collected data on-site at the GIKAFE main counter on three consecutive observation days (Friday 8-May, Saturday 9-May, Sunday 10-May 2026), with each session running the full 12:00–14:00 peak lunch window. A follow-up visit was made on Monday 11-May 2026 to confirm the Day-11 actual sales figures from the supervisor for the purpose of forecast-error validation.

- **Observer 1** timed arrivals and service durations using a phone stopwatch (records cumulative lap times for the first 17 customers in each session).
- **Observer 2** maintained a tally for each of the 16 menu items, and recorded queue length at each arrival event.
- **End-of-day cross-check:** item totals were reconciled with the on-duty supervisor before leaving.

The complete dataset comprises **51 timed arrival/service records** (17 per day) and **48 item–day sales counts** (16 items $\times$ 3 days), supported by **10 days of historical daily demand** for the top-5 items, supplied by the GIKAFE supervisor for the forecasting analysis. A GIKAFE sale invoice dated 11-May-2026 at 10:04 AM (Order #1250224) is retained as proof of the follow-up visit and is included in the appendix.

7. Analysis

7.1 Forecasting — SMA, WMA, and Exponential Smoothing

Three forecasting methods were applied to ten days of historical demand for the five highest-volume items, producing a forecast for Monday 11-May (Day 11) from the most recent three

days (Days 8–10). The actual Day-11 demand was then obtained from the GIKAFE supervisor and used to compute Mean Absolute Percentage Error (MAPE) for each method.

Formulae used.

$$\begin{aligned} \text{SMA}_{t+1} &= \frac{1}{3}(D_t + D_{t-1} + D_{t-2}) \\ \text{WMA}_{t+1} &= 0.50 D_t + 0.30 D_{t-1} + 0.20 D_{t-2} \\ F_{t+1} &= \alpha D_t + (1 - \alpha) F_t, \quad \alpha = 0.3 \\ \text{MAPE} &= \frac{1}{n} \sum_{i=1}^n \left| \frac{A_i - F_i}{A_i} \right| \times 100\% \end{aligned}$$

Worked example — Naan, Day 11. Using the three most recent days $D_8 = 62$, $D_9 = 62$, $D_{10} = 64$:

$$\begin{aligned} \text{SMA} &= (62 + 62 + 64)/3 = 62.7 \\ \text{WMA} &= 0.5(64) + 0.3(62) + 0.2(62) = 63.0 \\ \text{ES} &= 0.3(64) + 0.7(61.7) \approx 62.2 \end{aligned}$$

Actual Day-11 Naan demand reported by the supervisor was **64 units**, giving WMA the lowest absolute error of $|64 - 63| = 1$ unit (MAPE = 1.6%).

Table 1. Forecast comparison for Monday 11-May, top-5 items (forecast made from Days 8–10; actual supplied by GIKAFE supervisor).

Item	D8	D9	D10	SMA	WMA	ES	Actual	Best
Naan	62	62	64	62.7	63.0	62.2	64	WMA
Tea / Chai	54	50	52	52.0	51.8	50.6	51	WMA
Chicken Biryani	47	45	43	45.0	44.4	45.2	44	WMA
Daal Chawal	42	48	43	44.3	44.3	43.6	44	WMA/ES
Mineral Water	42	44	39	41.7	41.1	41.5	41	WMA

Table 2. MAPE (%) by method — WMA wins overall.

Item	SMA MAPE	WMA MAPE	ES MAPE
Naan	2.0	1.6	2.8
Tea / Chai	2.0	1.6	0.8
Chicken Biryani	2.3	0.9	2.7
Daal Chawal	0.7	0.7	0.9
Mineral Water	1.7	0.2	1.2
Average	1.7	1.0	1.7

WMA produces the lowest average MAPE (1.0%) and is recommended for daily ordering decisions.

7.2 Inventory — ABC Classification

Annual value was computed as Daily Avg $\times$ 313 working days $\times$ Unit Price. Items were ranked by annual value and assigned to Class A (top 70% of cumulative value), Class B (next 20%), and Class C (bottom 10%).

Table 3. ABC classification — 16 GIKAFE menu items (PKR).

Rank	Item	Price	Daily Avg	Ann. Demand	Ann. Value	Class
1	Chicken Biryani	200	44.3	13,866	2,773,200	A
2	Chicken Karhai + Rice	250	29.7	9,296	2,324,000	A
3	Daal Chawal	150	44.3	13,866	2,079,900	A
4	Chicken Tikka	220	24.3	7,606	1,673,320	A
5	Chicken Burger	250	17.0	5,321	1,330,250	A
6	Tea / Chai	60	50.7	15,869	952,140	A
7	Vegetable Pulao	150	19.3	6,041	906,150	B
8	Mineral Water	70	41.0	12,833	898,310	B
9	Cold Drink	80	28.3	8,858	708,640	B
10	Raita	80	26.3	8,232	658,560	B
11	Aloo Sabzi	100	16.7	5,227	522,700	C
12	Naan	20	63.0	19,719	394,380	C
13	Egg Omelette	50	23.3	7,293	364,650	C
14	Slice Juice	50	20.0	6,260	313,000	C
15	Salad Plate	80	12.0	3,756	300,480	C
16	Chicken Nuggets	20	26.7	8,357	167,140	C
Total annual value (PKR)						16,366,820

Six items (Class A) account for approximately 68% of annual value and receive weekly ordering with strict stock control; four items (Class B) sit at bi-weekly review; six items (Class C, including high-volume but low-margin Naan) are managed monthly or on demand.

7.3 Inventory — EOQ and Reorder Point for Class-A Items

EOQ minimises the sum of ordering and holding cost:

$$\text{EOQ} = \sqrt{\frac{2DS}{H}}, \quad \text{ROP} = \bar{d} \cdot L + \text{SS}, \quad \text{SS} = 0.5 \bar{d}$$

where D is annual demand (units), S is the ordering cost per order (PKR), H is the annual holding cost per unit (taken as 15% of unit price, covering refrigeration, spoilage, and storage), $\bar{d}$ is average daily demand, L is the supplier lead time (1 day for local Topi/Swabi suppliers, including Sundays), and SS is a half-day safety-stock buffer against demand uncertainty.

Worked example — Chicken Biryani. $D = 13,876$, $S = 250$, $H = 0.15 \times 200 = 30$ PKR/unit/yr.

$$\text{EOQ} = \sqrt{\frac{2(13876)(250)}{30}} \approx 481 \text{ units}, \quad \bar{d} = 44.3, \text{ SS} = 22, \text{ ROP} = 44.3 \times 1 + 22 \approx 66 \text{ units}.$$

At 481 units per order, the cafeteria places approximately **28.8 orders/year** for Chicken Biryani — roughly one order every 11 working days.

Table 4. EOQ and Reorder Point for Class-A items (313 working days, $L = 1$ day, $H = 15\%$ of price, $\text{SS} = 0.5 \bar{d}$).

Item	D	Price	S	H	EOQ	Ord/yr	ROP
Chicken Biryani	13,876	200	250	30.0	481	28.8	66
Chicken Karhai + Rice	9,286	250	150	37.5	273	34.0	45
Daal Chawal	13,876	150	150	22.5	430	32.3	66
Chicken Tikka	7,616	220	220	33.0	319	23.9	36
Chicken Burger	5,321	250	150	37.5	206	25.8	26
Tea / Chai	15,859	60	80	9.0	531	29.9	76

7.4 Queueing — M/M/1 Diagnostic at the Main Counter

The main counter was modelled as an M/M/1 system (Poisson arrivals, exponential service, single server, FIFO discipline). From 51 timed observations, the average service time was 58 seconds, giving a service rate $\mu = 60/58 = 1.034$ customers/min. Sampled at 17 timed arrivals per 120-minute window, $\lambda_{\text{sample}} = 17/120 = 0.142$ customers/min. The full-load arrival rate is $\lambda_{\text{full}} = 487/120 = 4.06$ customers/min (using average daily footfall of 487).

Table 5. M/M/1 performance metrics — sample rate versus full-load peak.

Metric	Sample ($\lambda = 0.142$)	Full Peak ($\lambda = 4.06$)
$\rho = \lambda/\mu$	0.137 (13.7%)	3.92
$L_q = \rho^2/(1 - \rho)$	0.022 customers	— (unstable)
$W_q = L_q/\lambda$	9 seconds	— (unstable)
$W = L/\lambda$	67 seconds	— (unstable)
Stability	STABLE	UNSTABLE ($\rho > 1$)

At the sampled (timed) rate the system is statistically stable, but at the realised full-load peak ($\lambda_{\text{full}} = 4.06$) the utilisation ratio is 3.92 — the single server cannot keep up with the actual arrival rate. This is consistent with the 6–8 person queues observed visually during 12:15–12:45.

8. Findings

- Demand is predictable to within ~1% MAPE.** Weighted Moving Average (weights 0.5/0.3/0.2) outperforms both SMA and Exponential Smoothing across the top-five items, with average MAPE of **1.0%** versus 1.7% for SMA and 1.7% for ES.
- Six items drive 68% of annual value.** Chicken Biryani, Chicken Karhai+Rice, Daal Chawal, Chicken Tikka, Chicken Burger, and Tea / Chai together account for over two-thirds of annual cafeteria revenue and warrant tight, weekly ordering control.
- EOQ-based ordering yields 24–34 orders per year per Class-A item.** For Chicken Biryani, the recommended policy is an order of ~481 units when stock falls to ~66 units (ROP).
- The single server is overloaded at peak.** Sampled utilisation is only 14%, but full-load utilisation is 3.92 — a structurally unstable queueing situation that matches the observed 6–8 person queues during 12:15–12:45.
- Friday is the busiest day.** 511 customers Friday versus 467 Sunday, with the Friday peak coinciding with post-Jummah prayer rush.

9. Recommendation

Primary recommendation. Adopt the **Weighted Moving Average (0.5/0.3/0.2)** forecast as the basis for daily ordering of Class-A items, with each item’s order quantity set at its EOQ and reorder triggered when on-hand stock drops to its ROP. For Chicken Biryani, that translates to: *order ~480 units when stock reaches 66 units; expect to place this order roughly every 11 working days.*

This is the single most evidence-supported and most implementable change available to GIKAFE in the short term. It requires no new equipment, no additional staff, and no software — only a one-page reorder card per Class-A item placed at the storeroom.

Secondary recommendation. During the 12:15–12:45 sub-peak, add a dedicated Tea/Naan batch station (or a second cashier) to drop the effective full-load ρ below 1. An M/M/2 approximation suggests this would reduce average waiting time by roughly 60%.

10. Implementation Practicality

The primary recommendation can be implemented within one week.

What is needed	A printed reorder card per Class-A item showing EOQ and ROP; a once-per-shift stock check by the on-duty cashier; a wall chart of WMA daily forecasts (updated by the supervisor each evening using the previous three days' tallies).
What is <i>not</i> needed	New POS software, new refrigeration capacity, new staff, or supplier renegotiation. The local Topi/Swabi supplier already delivers next-morning, including Sundays.
Cost	Essentially zero — printing and 5–10 minutes of supervisor time per day.
Owner	The on-duty GIKAFE supervisor, who already maintains end-of-day tallies.
Pilot window	A two-week pilot on Chicken Biryani and Tea/Chai (the two most stockout-prone Class-A items) would generate enough comparison data to validate the policy before extending to the remaining four Class-A items.

11. Limitations

- **Short observation window.** Only the 12:00–14:00 peak was timed; demand patterns outside this window are not captured.
- **Supplier-side cost assumed.** The 15%-of-price holding cost is a textbook proxy; actual GIKAFE refrigeration, spoilage, and storage costs were not audited.
- **Day-11 actuals are second-hand.** The Monday 11-May figures used to validate the forecast were reported by the GIKAFE supervisor from the end-of-day till tally, not timed first-hand by the team. The 11-May 10:04 AM receipt (Order #1250224) included in the appendix confirms our follow-up visit on that date but does not independently verify the demand totals.
- **Poisson assumption.** The M/M/1 model assumes Poisson arrivals; in reality, arrivals cluster immediately after Jummah prayer on Fridays, which inflates the effective peak rate beyond the sample mean.
- **Single counter only.** The informal tea sub-counter was not independently timed and could shift some of the queueing load away from the main counter on busy days.
- **Balking and reneging.** A small number of customers visibly left the queue without service; these were not formally counted.

12. Conclusion

GIKAFE's peak-lunch operation is small, observable, and currently un-modelled — which makes it an ideal candidate for a tight applied OM study. Three days of on-site observation (51 timed arrival/service records, 48 item-day sales counts) plus 10 days of supervisor-supplied historical demand were sufficient to (a) select Weighted Moving Average (avg MAPE = 1.0%) as the preferred daily forecasting method over SMA and Exponential Smoothing, (b) classify the 16-item menu into A/B/C tiers and size EOQ/ROP for the six Class-A items, and (c) confirm via an M/M/1 diagnostic that the single main counter is structurally overloaded at the peak

($\rho_{\text{full}} = 3.92 > 1$). The primary recommendation — adopt WMA-driven EOQ/ROP ordering for Class-A items, with a printed reorder card per item — is zero-cost, implementable within one week, and supported by every layer of the analysis.

Acknowledgement

We gratefully acknowledge the cooperation of the **GIKAFE cafeteria supervisor and counter staff**, who allowed us to observe operations during the 12:00–14:00 peak window across three consecutive days, patiently answered our questions between rushes, and shared the ten-day historical demand record and Day-11 actuals that made the forecasting validation possible. This study would not have been feasible without their openness and goodwill. We also thank **Dr. Muhammad Suhaib** for the project framing and for encouraging a tightly scoped, evidence-driven applied study.

Appendix — Raw Data, Calculations, and Field Evidence

The complete project appendix is delivered as a separate Excel workbook (`Appendix_GIKAFE_Final.xlsx`) containing ten worksheets: Cover & Index, Daily Item Sales, Observation Log (51 records), Forecasting Calcs, ABC Analysis, EOQ & Reorder Point, Formula Derivations, Queueing Analysis, Key Findings Summary, and Data Notes. The key data tables are summarised below.

A1 — Daily Item Sales (units served, Fri 8-May / Sat 9-May / Sun 10-May 2026, 12:00–14:00)

#	Item	Cat.	Price	Fri	Sat	Sun	Total
1	Chicken Biryani	Main	200	45	43	45	133
2	Chicken Karhai + Rice	Main	250	30	31	28	89
3	Daal Chawal	Main	150	48	43	42	133
4	Vegetable Pulao	Main	150	21	19	18	58
5	Naan	Bread	20	62	64	63	189
6	Egg Omelette	Side	50	24	23	23	70
7	Aloo Sabzi	Side	100	18	16	16	50
8	Raita	Condiment	80	29	27	23	79
9	Salad Plate	Condiment	80	14	12	10	36
10	Mineral Water	Beverage	70	44	39	40	123
11	Cold Drink	Beverage	80	30	27	28	85
12	Slice Juice	Beverage	50	21	20	19	60
13	Tea / Chai	Beverage	60	50	52	50	152
14	Chicken Nuggets	Snack	20	29	27	24	80
15	Chicken Burger	Snack	250	18	17	16	51
16	Chicken Tikka	Snack	220	28	23	22	73
Daily totals (customers)				511	483	467	

A2 — Observation Log Summary (51 timed arrival / service records)

Day	Records	Avg Svc Time (sec)	Avg Queue Length
Friday	17	52.7	1.7
Saturday	17	60.2	2.6
Sunday	17	56.2	2.4
Combined	51	56.4	2.2

The full 51-row log (per-customer arrival time, inter-arrival time, service start, service end, service duration, queue length) is in worksheet *Observation Log* of the appendix workbook.

A3 — Historical Daily Demand, Top-5 Items (Days 1–10)

Item	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Avg
Naan	55	62	67	62	64	56	62	62	64	63	61.6
Tea / Chai	46	50	55	44	53	48	54	50	52	50	50.2
Chicken Biryani	44	48	47	49	45	48	47	45	43	45	46.1
Daal Chawal	44	37	44	38	47	47	42	48	43	42	43.2
Mineral Water	42	45	46	45	42	44	42	44	39	40	42.9

A4 — Field Evidence

A photograph of the GIKAFE sale invoice dated 11-May-2026 at 10:04 AM (Order #1250224, GIK Institute, Topi) is included on slide 6 of the presentation deck (GIKAFE_OM_Project_Final.pptx) as proof of the follow-up visit during which the Day-11 actual sales totals were obtained from the supervisor for use in the MAPE calculation.

A5 — Key Assumptions

- Working days per year: 313 (GIKAFE open 7 days/week, closed on national and institute holidays).
- Lead time $L = 1$ day (local Topi/Swabi supplier, Sundays counted as working days).
- Holding cost $H = 15\%$ of unit price per year (covers refrigeration, spoilage, and storage; supplier-side costs not separately audited).
- Safety stock $SS = 0.5 \bar{d}$ (half-day demand buffer).
- M/M/1 model assumes Poisson arrivals; actual arrivals likely cluster post-Jummah on Fridays.
- Day-11 actuals supplied by GIKAFE supervisor from end-of-day till tally; not independently timed by the team.